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SUPERSEDING
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FEDERAL SPECIFICATION

CLOTH, COTTON, BUCKRAM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—This specification covers the requirements for dyed or bleached cloth, cotton, buckram, primarily intended for use in the manufacturing of binders for books and stiffening material for clothing (see 6.1).

1.2 **Classification.**

1.2.1 *Types and classes.*—The cloth, cotton, buckram shall be of the following types and classes, as specified (see 6.2).

Type I.—For bookbinding.
- Class 1.—4.7 ounces per square yard—light.
- Class 2.—5.7 ounces per square yard—medium.
- Class 3.—7.9 ounces per square yard—heavy.

Type II.—For clothing.
- Class 1.—3.8 ounces per square yard.
- Class 2.—4.5 ounces per square yard.
- Class 3.—12.0 ounces per square yard.
- Class 4.—17.0 ounces per square yard.

Type III.—For clothing, double texture (2 plies of type II, class 3).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CCC-T-191—Textile Test Methods.

PPP-P-51 — Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

FSC 8305

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(Copies of Military Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Standard sample. — Unless otherwise specified, the cloth shall be equal to the standard sample in all respects not specifically covered by this specification (see 6.3).

3.2 Preproduction sample approval.—Unless otherwise specified, before production is commenced the contractor shall submit for approval to the contracting officer, a 2-yard full-width sample of the cloth he proposes to furnish (see 6.2).

3.3 Material.

3.3.1 *Cotton.*—The cotton shall be of suitable length of staple and grade to meet the requirements of this specification.

3.3.2 *Yarn.* — The cotton used in the manufacture of the yarn shall be thoroughly cleaned, well carded, drawn and evenly spun.

3.4 Color. — The cloth shall be dyed or bleached, as specified (see 6.2). The color shall be produced by the use of such colorants and such methods as specified by the contracting officer. Sulfur dyes or dyes containing sulfur compounds susceptible to oxidation to sulfuric acid are prohibited (see 4.4.4).

3.4.1 *Matching.*—The color shall match an approved standard shade (see 6.3) under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin, and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin.

3.4.2 *Colorfastness.*

3.4.2.1 *Army, Navy, Marine Corps, and Civilian Government Agencies.*

3.4.2.1.1 The dyed cloth shall show fastness to water, crocking and light equal to or better than the standard sample, when tested as specified in 4.4.

3.4.2.1.1.1 When no standard sample is available or when the standard sample is not a reference for colorfastness the dyed cloth shall show good fastness to water and crocking and good fastness to light after 40 standard fading hours, when tested as specified in 4.4.

3.4.2.2 *Air Force.*—The fastness of the dyed cloth to light after 40 standard fading hours, to water, and to crocking shall be as specified by the contracting officer. Tests shall be conducted as specified in 4.4.

TABLE I.—*Physical requirements for type I*

Class	Width exclusive of selvage, minimum [1]	Weight per square yard		Number of yarns per inch, minimum		Breaking strength, minimum	
		Stripped [2] (minimum)	Finished (maximum)	Warp	Filling	Warp	Filling
	Inches	*Ounces*	*Ounces*			*Pounds*	*Pounds*
1	38	4.7	6.5	30	25	65	35
2	38	5.7	8.5	82	21	100	55
3	38	7.9	10.0	82	24	120	70

[1] Selvages shall not exceed ¼ inch.

[2] The basis of weight comparison shall be the weight of the fabric which remains after the finished fabric has been stripped of all starch and loading, or impregnating materials.

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3.5 Weave.—The weave shall be plain, except that type I, classes 2 and 3 shall be plain with 2 ends woven as one, and type II, class 3 shall be a 4 harness, 2 up and 2 down herringbone twill, 20 ends right and 20 ends left.

3.6 Type I, for bookbinding.

3.6.1 *Physical requirements.*—Type I cloth shall conform to the requirements in table I.

3.6.2 *Finish.*—The cloth shall have a starch finish.

3.6.3 *Length.*—Unless otherwise specified, type I cloth shall be furnished in rolls of one continuous piece of not less than 40 yards.

3.7 Type II, for clothing.

3.7.1 *Physical requirements.*—Type II cloth shall conform to the requirements in table II.

3.7.2 *Finish.*

3.7.2.1 *Class 1.*—Class 1 cloth shall have a swing finish produced by a shifting tenter frame which breaks the adhesion of the finish at the interstices. Determination of the swing finish shall be made by test specified in 4.4.3.

3.7.2.2 *Class 2.*—Class 2 cloth shall have a smooth permanent finish suitable for lining shirt collars. The material shall show no discoloration, chlorination, degradation, or loss of more than 10 percent of the original breaking strength of the fabric when tested as specified in 4.4.2.

3.7.2.3 *Classes 3 and 4.*—Classes 3 and 4 cloths shall have a starch finish.

3.7.3 *Shrinkage.*—Type II, class 2 cloth shall not shrink more than one percent in either the warp or filling direction when tested as shown in table IV. The preshrinking process used shall not be identified by name or trade mark, either on the cloth, ticket, or the package.

3.7.4 *Length.*—Unless otherwise specified, type II cloth shall be furnished in rolls of one continuous piece of not less than 50 yards.

3.8 Type III, double texture for clothing.

3.8.1 *Construction.*—Type III cloth shall be made of two plies of type II, class 3 cloth and shall be joined by laminating the two plies together.

3.8.2 *Laminating material.*—The laminating material shall be of a synthetic resin type, unaffected by standard dry cleaning solvents and standard launderings. There shall be no objectionable odor caused by these materials.

3.8.2.1 *Thickness.*—The thickness of the laminating material shall be not less than 0.014 inch nor more than 0.016 inch.

TABLE II.—*Physical requirements for type II*

Class	Width inclusive of selvage, minimum [1]	Finished weight per square yard, minimum	Number of yarns per inch, minimum		Breaking strength, minimum		Sizing and other non-fibrous materials, maximum
			Warp	Filling	Warp	Filling	Percent
	Inches	*Ounces*			*Pounds*	*Pounds*	
1	35½	3.8	46	42	50	40	35
2	32	4.5	50	42	60	50	60
3	34	12.0	64	20	150	80	50
4	34	17.0	74	21	130	40	65

[1] Selvages shall not exceed ¼ inch.

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3.8.3 *Breaking strength.* — The breaking strength of the laminated cloth shall be not less than that required for a single ply of type II, class 3 cloth when tested as specified in table IV.

3.8.4 *Stiffness.*—Type III cloth shall have an average stiffness of not less than 0.400 inch-pounds at 60 degrees angular deflection for both warp and filling when tested as specified in table IV.

3.8.5 *Resistance to dry cleaning.* — The laminated cloth shall show no evidence of delamination when subjected to three cycles of dry cleaning when tested as specified in table IV.

3.8.6 *Resistance to laundering.*—The laminated cloth shall show no evidence of delamination when subjected to six cycles of laundering when tested as specified in table IV.

3.8.7 *Length.*—Unless otherwise specified, type III cloth shall be furnished in rolls of one continuous piece of not less than 40 yards.

3.9 Identification tickets.—A ticket and the attachment to each roll shall be in accordance with identification requirements of Federal Specification PPP–P–51. Except when otherwise specified by Civilian Government Agencies the following information shall be printed on the ticket:

Nomenclature
Type_____, Class_____,
Color_____, Width_____
Stock number
Contractor's name
Contractor's piece number
Contract number and date
Specification number and date
Name of contracting office
Gross yards
Net yards

In addition, the ticket shall have sufficient blank space for the inspector's stamp.

3.10 Workmanship.—The cloth shall conform to the quality and grade of product established by this specification, and the occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.—The quality assurance provisions contained in this section are applicable to inspection. Quality assurance provisions for components or materials described shall be in accordance with this specification and with subsidiary specifications referenced herein to the extent applicable, except that this specification shall govern in the event of conflict.

4.2 Pre-acceptance inspection.

4.2.1 *Preproduction sample inspection.*—The preproduction sample submitted in accordance with 3.2 shall be visually inspected for appearance and finish. The sample shall be tested for physical and chemical properties in accordance with the methods shown in table IV, 4.4.1, 4.4.2, 4.4.3 and 4.4.4.

4.3 Inspection for acceptance. — Except when otherwise specified by Civilian Government Agencies, inspection shall be in accordance with the provisions of Military Standard MIL–STD–105 except where otherwise indicated.

4.3.1 *Inspection of the end item.*

4.3.1.1 *Examination of the end item.*—Defects found during this examination shall be classified in accordance with the list shown in 4.3.1.1.1, 4.3.1.1.2 and 4.3.1.1.3.

4.3.1.1.1 *Yard by yard examination.*—The required yardage of each piece shall be inspected and visual defects classified as listed. The defects found shall be counted, regardless of their proximity to each other, except where two or more defects represent a single condition of the cloth in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this

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examination shall be 1 linear yard. The acceptable quality levels (AQL'S) for this examination shall be 1.5 major defects and 6.5 total defects (majors and minors combined) per 100 units (yards). The sample size shall be based upon inspection level II, table III of Military Standard MIL–STD–105. The lot size shall be expressed in units of 1 yard each. The number of pieces from which the sample is to be selected shall be in accordance with table III. An approximate equal number of yards shall be examined in each piece of the sample.

Glossary No. Fed. Std. No. 4	Defects	Classification	
		Major	Minor
90, 91, 92	Hole, cut or tear—three or more yarns ruptured in body of cloth	X	
78	Smash, any	X	
11, 33	Broken or missing yarns		
	Three or more contiguous regardless of length	X	
	Two contiguous, 2 inches or more missing	X	
	Two contiguous, less than 2 inches missing or single, 2 inches or more missing		X
9	Fine filling bar or light place—clearly noticeable	X	
9, 12, 35	Open place, crack or hitchback—opening at widest point four or more normal yarn diameters or sequence of openings resulting in a weak place	X	
13, 34	Floats or skips		
	Multiple, ½ inch or more in combined warp and filling directions, or harness skips	X	
	Multiple less than ½ inch in combined warp and filling directions, or single floating 2 inches or more		X
1, 2, 4, 10, 26	Heavy filling bar, heavy place, coarse or additional yarns—clearly noticeable	X	
3, 5, 27, 77	Slough off, jerked-in filling, slugs, slubs or gouts—three or more times the thickness of normal yarn	X	
6, 7, 28, 29	Knots or kinks, large—forming hard lumps	X	
55, 56	Selvage cut, torn or broken—extending more than ½ inch from edge of cloth	X	
49, 50	Selvage slack or tight—clearly noticeable		X
48, 51	Selvage folded, rolled, curled or scalloped, width of cloth inside defect less than specified		X
89	Tender or weak place	X	
69, 70, 74, 75, 76	Spot, stain or streak—uncleaned	X	
59, 61	Crease or wrinkle—hard embedded		X

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4.3.1.1.2 *Over-all examination.*—Each defect listed below shall be counted not more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size (number of pieces selected as sample) for this examination and the number of defects acceptable shall be as specified in table III.

Defects

Over-all uncleanness, objectionable odor
Width less than specified
Uneven weaving throughout the piece
Shade (when colors are specified)
Off shade, shaded end to end, side to side, side to center or throughout the piece
Cloudy, mottled or streaky throughout the piece.

TABLE III.

Lot size in yards	Sample size in pieces	Maximum number of defects acceptable in sample
Up to 1,300 [1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	1
110,001 to 550,000	25	1
550,001 and over	50	2

[1] If a lot contains less than three pieces, each piece in the lot shall be examined.

4.3.1.1.3 *Examination for length.*

4.3.1.1.3.1 *Individual pieces.* — The piece shall be examined for gross length. Any gross length found to be less than the minimum specified, or more than 2 yards below the gross length marked on the ticket, shall be considered a defect with respect to length. The unit of product for this examination shall be one piece. The number of pieces selected as sample (sample size) for this examination and the acceptance number shall be in accordance with table III.

4.3.1.1.3.2 *Total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross lengths of pieces in the sample is less than the total of the gross lengths marked on the tickets.

4.4 Testing of the end item.—The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in 4.4.1, 4.4.2, 4.4.3 and 4.4.4 shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be two continuous yards full width of the finished fabric. For each characteristic, the acceptable quality level shall be 4.0 test failures per 100 units (yards) and the inspection level shall be level L–1, table VII of Appendix to Military Standard MIL–STD–105. The lot size shall be expressed in units of 1 yard each.

TABLE IV.—*Test methods*

Characteristic	Requirement reference	Test method
Color fastness to:		
Light	3.4.1	5660
Water	3.4.1	5630
Crocking	3.4.1	5650
Weave	3.5	Visual
Weight, oz. sq. yd.	Tables I and II	5041
Yarns per inch	Tables 1 and II	5050
Breaking strength	Tables I and II, 3.8.3	(see 4.4.1)
Non-fibrous materials	Table II	2611
Swing finish	3.7.2.1	4.4.3
Chlorine degradation	3.7.2.2	4.4.2
Shrinkage in laundering	3.7.3	5550
Thickness	3.8.2.1	5030
Stiffness	3.8.4	5202
Dry cleaning (3 cycles)	3.8.5	5622
Laundering (6 cycles)	3.8.6	5550

4.4.1 *Breaking strength.*

4.4.1.1 Type I, classes 1, 2 and 3, and type II, class 4, shall be tested for breaking

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strength in accordance with method 5102 of Federal Specification CCC–T–191, except that the specimens shall be obtained as follows: A strip of cloth 7 inches long in the warp direction shall be cut from across the entire width of the cloth. Twelve specimens shall be cut from this strip, six with the long dimension in the direction of the warp and six with the long dimension in the direction of the filling. Each specimen shall be 1 inch in width. No two specimens shall contain the same yarns and care should be taken to insure that the yarns to be subjected to test have not been damaged during the cutting of the specimens.

4.4.1.2 Type II, classes 1, 2 and 3 shall be tested for breaking strength in accordance with method 5100 of Federal Specification CCC–T–191.

4.4.2 *Test for deleterious effect of finish on breaking strength (type II, class 2).* — A piece of cloth, not less than 20 inches long and full width shall be laundered in accordance with method 5550 of Federal Specification CCC–T–191 and shall then be treated for one hour with 0.7 percent available chlorine solution. The piece shall be rinsed in a sour solution and washed thoroughly until a control piece shows no trace of chlorine. The piece shall be air dried and then conditioned for 4 hours in an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of 75° ± 5° F. Eight pairs of specimens, four warpwise and four fillingwise, with each pair having the same yarns shall be taken for tests. Four warp specimens and four filling specimens shall be tested in a pendulum type machine in accordance with method 5100 of Federal Specifications CCC–T–191. The remaining warp and filling specimens shall be oven dried at 105° to 110° C. for two hours and tested in the same manner as the first eight specimens. The differences in the average breaking strength of the respective warp and filling sample shall be computed and the percentage of loss determined. If a specimen slips or breaks in or at the edge of the jaws, the result shall be discarded and another pair of specimens taken, both having identical warp and filling yarns and the test repeated on this pair of specimens. (See 3.7.2.2.)

4.4.3 *Swing finish.* — A piece of type II, class 1 cloth 10 by 10 inches shall be folded on a 45-degree bias without creasing. Holding the ends of the fold with both hands and exerting a pull the cloth will stretch or bulge. Cloth having no swing finish will not stretch or buckle.

4.4.4 *Test for sulfur dyes.* — Sulfur dyes are determined by tests for available sulfur.

4.4.4.1 *Procedure.*—Dissolve 100 grams of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water. One to 1.5 gram of the finished fabric, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with the stannous chloride solution. Place a filter paper which has been moistened with one drop of lead acetate solution, over the mouth of the tube. (Punch needle hole in paper.) Heat the test tube slowly over a low flame until the solution is boiling. (Prolonged boiling will reduce the sensitivity of the test.) A brown stain appearing on the paper indicates the presence of sulfur.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Levels A, B, and C.* — The rolls of cloth shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51.

5.2 Packing.

5.2.1 *Levels A, B, and C.*—The rolls of cloth, packaged as specified in 5.1.1, shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

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5.3 Marking.—Shipments shall be marked in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.4 Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6. NOTES

6.1 Intended use.—Type I cloth (for bookbinding) is used for textbook, edition, trade, catalog, and similar uses, except that it should not be used for library bindings. Type II, class 2 (for clothing) is suitable for lining shirt collars.

6.2 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of cloth required (see 1.2).
(*c*) When the cloth need not be equal to the standard sample (3.1).
(*d*) When preproduction sample approval is not required (3.2).
(*e*) Color (3.4).
(*f*) Length (3.6.3, 3.7.4, and 3.8.7).
(*g*) Identification tag—legend (3.9).
(*h*) Inspection (4.3).
(*i*) Both the level of packaging and the level of packing required (5.1, 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in PPP–P–51 may be exercised in the preparation for delivery of this commodity.

6.3 Standard sample.—For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Suggested basic cloths for type I buckram are as follows:

Class 1—42 inches, 36 by 32, 2.60 yards per lb. clear Osnaburg.
Class 2—41 inches, 84 by 26, 2 yards per lb. SF Flat Duck.
Class 3—42 inches, 84 by 28, 1.45 yards per lb. DF Flat Duck.

6.5 Transportation description.—The transportation description applicable to this item is:

Buckram, cotton, single or other than single ply.

Carload minimum weight, any quantity.

Buckram, cotton. (Indicate single or other than single ply.)

Motor volume minimum weight, any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—Q
Navy—S MC S H
Air Force.

Copies of this specification may be purchased for 5 cents each.

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